

LINEAR NEURON

LOGISTIC NEURON

MULTI-LAYER NETWORKS

LINEAR NEURON

①

$$W = \{ \omega_0, \omega_1, \omega_2, \dots \}^T$$

$$X = \{ 1, x_1, x_2, \dots \}^T$$

$$y = W^T X = \sum_j \omega_j x_j$$

$$E = \frac{1}{2} \sum_i (t^i - y^i)^2$$

$$E = \frac{1}{2} \sum_i \left(t^i - \sum_j \omega_j x_j^i \right)^2$$

$$\frac{\partial E}{\partial \omega_j} = \frac{\partial}{\partial \omega_j} \frac{1}{2} \sum_i \left(t^i - \sum_j \omega_j x_j^i \right)^2$$

$$= \sum_i \left(t^i - \sum_j \omega_j x_j^i \right) x_j^i$$

$$\boxed{\frac{\partial E}{\partial \omega_j} = - \sum_i (t^i - y^i) x_j^i}$$

ITERATIVE SOLUTION

$$\omega_j \leftarrow \omega_j - \eta \frac{\partial E}{\partial \omega_j}$$

$$\Delta \omega_j = -\eta \frac{\partial E}{\partial \omega_j} = \eta \sum_i (t^i - y^i) x_j^i$$

$$\omega_j \leftarrow \omega_j + \Delta \omega_j$$

DELTA RULE

$$E = \frac{1}{2} \sum_{i \in \text{train}} (t^i - y^i)^2$$

$$\frac{\partial E}{\partial w_j} = \frac{1}{2} \sum_i \frac{\partial y^i}{\partial w_j} \frac{\partial E^i}{\partial y^i}$$

Chain rule

$$\frac{\partial E}{\partial w_j} = - \sum_i x_j^i (t^i - y^i)$$

$$\Delta w_j = -\eta \frac{\partial E}{\partial w_j} = \eta \sum x_j^i (t^i - y^i)$$

TOY EXAMPLE

x_f : quantity or portion of fish

x_c : " " chips

x_k : " " Ketchup

w_f : price of fish

w_c : " " chips

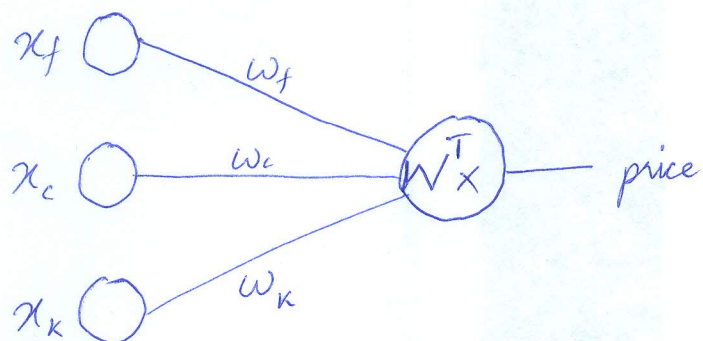
w_k : " " Ketchup

$$W = \{w_f, w_c, w_k\}^T$$

$$X = \{x_f, x_c, x_k\}^T$$

$$\text{total price} = x_f w_f + x_c w_c + x_k w_k = W^T X$$

③



Receipt 1	
Fish	2
Chips	5
Ketchup	3
Total Price	850

Let randomly initialize $W = \{50, 50, 50\}$

$$\text{total price} = 2 \times 50 + 5 \times 50 + 3 \times 50 = 500$$

$$\Delta w_j = \eta x^i (t - y)$$

$$\text{Let } \eta = \frac{1}{35}$$

$$\therefore \Delta w_f = \frac{1}{35} 2 (850 - 500) = 20$$

$$\Delta w_c = \frac{1}{35} 5 (850 - 500) = 50$$

$$\Delta w_k = \frac{1}{35} 3 (850 - 500) = 30$$

$$\therefore W = W + \Delta W \quad \text{or} \quad w_j = w_j + \Delta w_j$$

$$\therefore w_f = 50 + 20 = 70$$

$$w_c = 50 + 50 = 100$$

$$w_k = 50 + 30 = 80$$

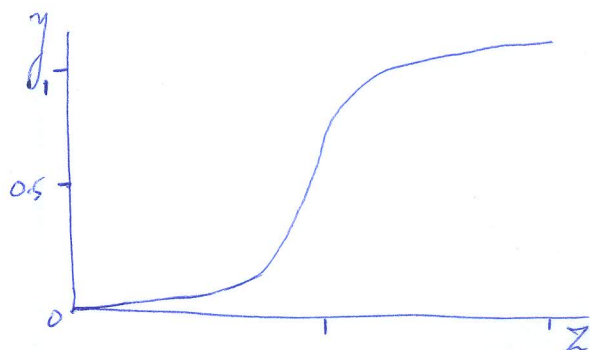
$$W = \{70, 100, 80\}$$

LOGISTIC NEURON

④

$$z = \sum_j w_j x_j = W^T X$$

$$y = \frac{1}{1 + e^{-z}}$$



$$\frac{\partial z}{\partial w_j} = x_j$$

$$\frac{\partial z}{\partial x_j} = w_j$$

$$\frac{\partial y}{\partial w_j} = \frac{\partial z}{\partial w_j} \frac{\partial y}{\partial z}$$

$$\frac{\partial y}{\partial z} = \frac{\partial}{\partial z} \left(\frac{1}{1 + e^{-z}} \right)$$

$$= \frac{\partial}{\partial z} (1 + e^{-z})^{-1}$$

$$= -(1 + e^{-z})^{-2} \frac{\partial}{\partial z} (1 + e^{-z})$$

$$= -(1 + e^{-z})^{-2} \cdot e^{-z} \frac{\partial}{\partial z} (-z)$$

$$= \frac{e^{-z}}{(1 + e^{-z})^2} = \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} = \frac{1}{1 + e^{-z}} \cdot \frac{1 + e^{-z} - 1}{1 + e^{-z}}$$

$$\frac{\partial y}{\partial z} = \frac{1}{1+e^{-z}} \cdot \left(\frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}} \right)$$

(5)

$$= y(1-y)$$

$$\therefore \frac{\partial y}{\partial w_j} = \frac{\partial z}{\partial w_j} \frac{\partial y}{\partial z} = x_j y(1-y)$$

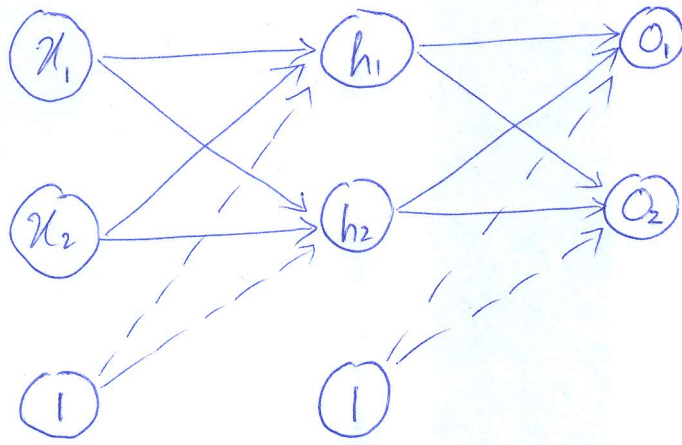
$$\frac{\partial E^i}{\partial w_j} = \sum_i \frac{\partial z^i}{\partial w_j} \frac{\partial y^i}{\partial z^i} \frac{\partial E^i}{\partial y^i}$$

$$\frac{\partial E^i}{\partial w_j} = - \sum_i x_j^i y^i (1-y^i) (t^i - y^i)$$

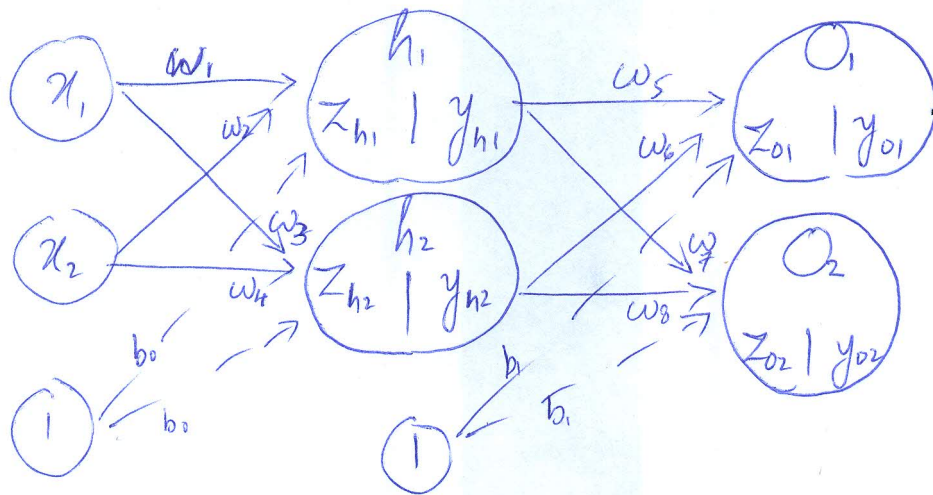
$$\therefore \Delta w_j = -\eta \frac{\partial E}{\partial w_j} = \eta \sum_i x_j^i y^i (1-y^i) (t^i - y^i)$$

MULTI-LAYER PERCEPTRON

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OR



$$\frac{\partial E}{\partial y} = -(t - y)$$

$$\frac{\partial y}{\partial z} = y(1 - y)$$

$$\frac{\partial z}{\partial w_j} = x_j$$

$$\frac{\partial z}{\partial x_j} = w_j$$

FORWARD PASS

⑦

$$Z_{h_1} = x_1 w_1 + x_2 w_2 + b_0 \quad \dots \quad Z_{h_1}$$

$$Z_{h_2} = x_1 w_3 + x_2 w_4 + b_0 \quad \dots \quad Z_{h_2}$$

$$y_{h_1} = \frac{1}{1 + e^{-Z_{h_1}}}$$

$$y_{h_2} = \frac{1}{1 + e^{-Z_{h_2}}}$$

$$Z_{o_1} = y_{h_1} w_5 + y_{h_2} w_6 + b_1$$

$$Z_{o_2} = y_{h_1} w_7 + y_{h_2} w_8 + b_1$$

$$y_{o_1} = \frac{1}{1 + e^{-Z_{o_1}}}$$

$$y_{o_2} = \frac{1}{1 + e^{-Z_{o_2}}}$$

$$E_{o_1} = \frac{1}{2} (t_{o_1} - y_{o_1})^2$$

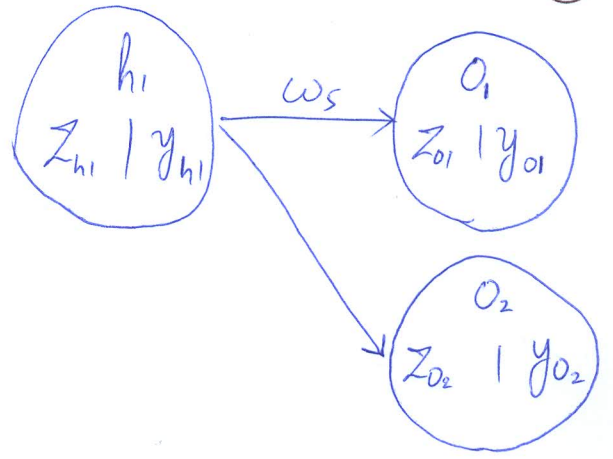
$$E_{o_2} = \frac{1}{2} (t_{o_2} - y_{o_2})^2$$

$$E = E_{o_1} + E_{o_2}$$

BACKWARD PASS

⑧

$$\frac{\partial E}{\partial w_5} = ?$$



$$\frac{\partial E}{\partial w_5} = \underbrace{\frac{\partial z_{o1}}{\partial w_5}}_{(3)} \underbrace{\frac{\partial y_{o1}}{\partial z_{o1}}}_{(2)} \underbrace{\frac{\partial E}{\partial y_{o1}}}_{(1)}$$

① $\frac{\partial E}{\partial y_{o1}} = \frac{\partial}{\partial y_{o1}} (E_{o1} + E_{o2})$

$$= \frac{\partial}{\partial y_{o1}} \frac{1}{2} (t_{o1} - y_{o1})^2 + \frac{\partial}{\partial y_{o1}} \frac{1}{2} (t_{o2} - y_{o2})^2$$

$$= -(t_{o1} - y_{o1}) + 0$$

$$= -(t_{o1} - y_{o1})$$

② $\frac{\partial y_{o1}}{\partial z_{o1}} = \frac{\partial}{\partial z_{o1}} \left(\frac{1}{1 + e^{-z_{o1}}} \right) =$

$$\therefore \frac{\partial y}{\partial z} = y(1-y)$$

$$\frac{\partial y_{o1}}{\partial z_{o1}} = y_{o1}(1-y_{o1})$$

$$\textcircled{3} \quad \frac{\partial z_{o1}}{\partial w_5} = \frac{\partial}{\partial w_5} (y_{h1} w_5 + y_{h2} w_6 + b_1)$$

$$= y_{h1}$$

$$\therefore \frac{\partial E}{\partial w_5} = -y_{h1} y_{o1} (1 - y_{o1})(t_{o1} - y_{o1})$$

$$\frac{\partial E}{\partial w_1} = ?$$

$$\frac{\partial E}{\partial w_1} = \frac{\partial z_{h1}}{\partial w_1} \frac{\partial y_{h1}}{\partial z_{h1}} \frac{\partial E}{\partial y_{h1}}$$

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②
①

$$\textcircled{1} \quad \frac{\partial E}{\partial y_{h1}} = \frac{\partial}{\partial y_{h1}} (E_{o1} + E_{o2})$$

$$= \frac{\partial E_{o1}}{\partial y_{h1}} + \frac{\partial E_{o2}}{\partial y_{h1}}$$

$$= \frac{\partial z_{o1}}{\partial y_{h1}} \cdot \frac{\partial y_{o1}}{\partial z_{o1}} \frac{\partial E_{o1}}{\partial y_{o1}} + \frac{\partial z_{o2}}{\partial y_{h1}} \frac{\partial y_{o2}}{\partial z_{o2}} \frac{\partial E_{o2}}{\partial y_{o2}}$$

$$\frac{\partial E}{\partial y_{h1}} = \frac{\partial (y_{h1} w_5 + y_{h2} w_6 + b_1)}{\partial y_{h1}} \frac{\partial}{\partial z_{o1}} \left(\frac{1}{1+e^{-z_{o1}}} \right) \cdot \frac{\partial}{\partial y_{o1}} \frac{1}{2} (t_{o1} - y_{o1})^2$$

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$$+ \frac{\partial (y_{h1} w_7 + y_{h2} w_8 + b_2)}{\partial y_{h1}} \frac{\partial}{\partial z_{o2}} \left(\frac{1}{1+e^{-z_{o2}}} \right) \frac{\partial}{\partial y_{o2}} \frac{1}{2} (t_{o2} - y_{o2})^2$$

$$\therefore \frac{\partial z}{\partial y} = y(1-y)$$

$$\frac{\partial E}{\partial y} = - (t - y)$$

$$\therefore \frac{\partial E}{\partial y_{h1}} = -w_5 y_{o1} (1 - y_{o1}) (t_{o1} - y_{o1}) - w_7 y_{o2} (1 - y_{o2}) (t_{o2} - y_{o2})$$

$$\textcircled{2} \quad \frac{\partial y_{h1}}{\partial z_{h1}} = \frac{\partial}{\partial z_{h1}} \frac{1}{1+e^{-z_{h1}}} = y_{h1} (1 - y_{h1})$$

$$\textcircled{3} \quad \frac{\partial z_{h1}}{\partial w_1} = \frac{\partial}{\partial w_1} (x_1 w_1 + x_2 w_2 + b_0) = x_1$$

$$\therefore \frac{\partial E}{\partial w_1} = x_1 y_{h1} (1 - y_{h1}) \left(-w_5 y_{o1} (1 - y_{o1}) (t_{o1} - y_{o1}) - w_7 y_{o2} (1 - y_{o2}) (t_{o2} - y_{o2}) \right)$$